

Pixel Art Is a Palette, Not a Downscale — Supplementary Material

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001	1. What the body left out		
002	The body reports what a reader needs to use the method.		
003	This document carries the rest: the data pipeline in enough		
004	detail to rebuild the corpus, the results that did not earn a		
005	place in the recipe, and the measurements behind claims the		
006	body states in one sentence. Section numbers here are inde-		
007	pendent of the body.		
008	2. Corpus construction		
009	From sheets to sprites. Source material arrives as sprite		
010	sheets and loose PNGs. Whole images whose longest side		
011	is at most 64 px are kept as they are — these are the na-		
012	tively tiny sprites the low-resolution buckets are starving		
013	for. Larger images are treated as sheets and cut into alpha-		
014	connected components between 8 and 64 px. Every cut		
015	records the archive, the member file and the licence it came		
016	with.		
017	Filters. A candidate is discarded unless its shorter side is		
018	at least 10 px, its aspect ratio is at most 3:1, the fraction		
019	of pixels with $\alpha > 16$ lies in $[0.15, 0.97]$, it has at least		
020	four distinct colours after quantising each channel to 8 lev-		
021	els, and the standard deviation of its opaque RGB is at least		
022	18. These reject background patches, single-colour blocks,		
023	and the near-empty crops that connected-component cutting		
024	produces at sheet margins.		
025	Deduplication. Every surviving sprite is hashed by the		
026	exact bytes of its RGBA array together with its shape. Within		
027	the original corpus this finds 4,888 extra copies. Because		
028	hold-out sets are defined by path, every copy of a held-		
029	out sprite is also excluded from training, not only the copy		
030	that was drawn. Near-duplicates — animation frames of the		
031	same character — are not removed from training: an abla-		
032	tion that collapsed them by an 8×8 luminance-plus-alpha		
033	average hash lost more data than it gained diversity (training		
034	rows 64,990 to 49,984) and scored worse at equal steps, so		
035	only the per-source cap below is applied.		
	Bucket policy. A sprite may feed a resolution bucket only	036	
	if its native size is at most $1.5\times$ that bucket. This is the	037	
	change that stops 25–64 px artwork from becoming a 16 px	038	
	target. It costs training rows: in the enlarged corpus, the	039	
	one whose training log we still have, it refuses 154,312 of	040	
	272,690 candidate (sprite, bucket) pairs. It is the single	041	
	largest contributor to the data-side gain.	042	
	Added material. 14,999 sprites cut from three permis-	043	
	sively licensed OpenGameArt collections (CC-BY-4.0, CC-	044	
	BY-SA-3.0, OGA-BY-3.0) pass the filters and cross-corpus	045	
	deduplication. A second pass drops sheets whose member	046	
	name matches a font, UI or text pattern (367 sprites), col-	047	
	lapses near-identical animation frames within a sheet, and	048	
	caps any single source sheet at 60 sprites so one anima-	049	
	tion cannot dominate a bucket. The survivors have a median	050	
	palette of eight colours and 6,028 of them are natively 24 px	051	
	or smaller.	052	
	Captions. Sprites are captioned by a vision–language	053	
	model shown numbered grids of 20 sprites on a checker-	054	
	board background, which is an order of magnitude cheaper	055	
	than one call per sprite. A text-only classifier then labels	056	
	each caption object / effect / fragment / glyph / junk; on	057	
	the 2,980 sprites that also carry image-based labels the two	058	
	agree 82.5% of the time (90.5% recall on objects), which is	059	
	enough to use the labels for an ablation and not enough to	060	
	filter the training corpus with them.	061	
	Licences. The sprites added in this work carry per-file	062	
	records of archive, member and licence. The older mate-	063	
	rial predates that bookkeeping; we release the manifests for	064	
	what we added and the scripts that rebuild everything else	065	
	from public archives.	066	
	3. Data interventions that did not earn a place	067	
	Three attempts to fix the corpus at training time. None is in	068	
	the final recipe; the second and third are reported because of	069	
	how they fail.	070	

Content filtering. A text classifier labels every caption object / effect / fragment / glyph / junk. Dropping the 8,818 sprites it calls fragment, glyph or junk (training rows 86,290 to 73,845) improves a 20k-step run from 165.6 to 156.7 native FD at equal steps, a 5.4% gain. That is real but small next to the sampler, it costs a full retraining, and the classifier itself agrees with image-based labels only 82.5% of the time, so we keep the unfiltered model.

Quality conditioning, and what it does to FD. A vision model rates each sprite’s craft from 1 to 5; on the original corpus 25.2% are rated 1–2 and 8.3% are rated 5. Prepending the rating to the caption as a tag and asking for the top tag at sampling time is the standard aesthetic-tag recipe and needs no architecture change. It works as intended in the only sense a person can check: the samples are visibly more finished. Both distributional scores get worse — 158.4 to 208.4 against the native reference, and 186.5 to 227.5 against a reference restricted to the 719 sprites rated 4–5 — because conditioning on the best 8% of the corpus narrows the output distribution, and FD measures distance to a distribution rather than to its best part. We keep the main model unconditioned and report the knob, together with the disagreement: a reader who looks at the samples and a reader who reads the table draw opposite conclusions, and both are reading their instrument correctly.

More data. The body reports the headline of this one: two additions that look decisive at 20k steps (54.2 to 42.9 on the reported configuration) and lose at 80k (39.3 $\pm$ 0.3 against 31.8 $\pm$ 0.4 over three seeds each). Table 1 gives both pilots and the full-length runs together. The two sources differ in what they add rather than how much: the Kenney packs are flatter than anything else we have but only 41.8% whole objects, while the CC-BY-3.0 bundle is 82.4% whole objects. At 20k the second helps ten times as much as the first, which invited the reading that flatness is what the decoder already supplies and shape is what it cannot. At 80k the effect is gone, and we do not offer a mechanism: the natively-small fraction of the added material is 24.0% against 28.6% for what was already there, nowhere near enough to explain a 24% regression.

Table 1. The enlarged corpus at two training lengths, 16 px native FD. The 20k column ranks the corpora one way and the 80k column the other.

Training corpus	20k steps	80k steps
Base (86,290 rows)	54.2	31.8 $\pm$ 0.4
+ Kenney (98,590)	53.2	—
+ Kenney + CC-BY-3.0 (118,378)	42.9	39.3 $\pm$ 0.3

4. Cross-resolution self-guidance: the negative result in full

A bucketed model offers a guidance reference with no extra weights: query the same network under a *lower* resolution label and extrapolate away from that prediction, $\tilde{e} = e_{\text{low}} + w(e_R - e_{\text{low}})$ with $e_{\text{low}} = \epsilon_{\theta}(x_t, t, c, b_{12})$. The reference is a prediction of what the same sprite would look like drawn smaller, so the update pushes away from “too coarse” rather than away from “unconditional”. We also trained a one-pass version, which internalises the rule into the weights with a guidance-free objective so that a single forward pass produces the guided prediction.

Under the conventional reference this was our strongest result. Under the native reference it is worth nothing, and the internalised version is actively harmful (Table 2). The two columns disagree on every row and they disagree in a consistent direction: the biased reference prefers more guidance, the native one prefers less.

Table 2. 16 px, main model, 3,000 captions. Both references, same samples. The biased column orders the rows almost exactly opposite to the native one.

Sampler	Biased $\downarrow$	Native $\downarrow$
CFG 1.0 (no guidance)	19.0	113.2
CFG 1.25	15.2	113.2
CFG 1.5	13.6	117.9
CFG 2	12.2	131.1
cross-resolution, $w=1.5$	14.2	114.9
composed reference, $w=1.5$	12.8	117.0
internalised, one pass	10.4	170.3
ICG $w=1.25$	15.5	94.2
ICG $w=1.5$	16.2	87.2
ICG $w=2$	20.7	87.1

The mechanism is visible in the per-image statistics. Guidance away from a lower-resolution prediction makes a sample look like a sharpened version of a downsampled image, which is exactly what the biased reference is made of, so the biased score improves monotonically with guidance weight. It does not reduce the colour count or increase flatness, which is what separates the native reference from everything else, so the native score does not move. Independent-condition guidance, which we did not invent, does help under both — and it composes with palette projection, which is why the final recipe uses it.

We report this at length because the two references would have led us to publish different papers. Had we scored only against the conventional reference, the internalised one-pass rule — 10.4, the best number in the table — would have been the contribution, and it is the single worst sampler in the column that measures distance to real pixel art.

146	5. Ablations on the model rather than the de-	
147	coder	
148	Before concluding that the remaining error is in the data and	
149	the decoder, we checked the three obvious properties of the	
150	network. None of them explains it.	
151	Text encoder. Swapping the frozen CLIP ViT-B/32 for	
152	ViT-L/14 (512 to 768 channels of cross-attention), same	
153	data, recipe and step count: 164.0 against 165.6 native FD at	
154	20k steps, a difference far inside what we can attribute, and	
155	caption retrieval goes slightly <i>down</i> , 20.6% to 19.6%. What-	
156	ever makes our samples draw the wrong object, it is not the	
157	text tower, and a full retraining with the larger encoder is not	
158	worth 30 GPU-hours.	
159	Training length. Continuing the main model past 80k	
160	steps flattens: 31.8 at 80k, 30.8 at 100k, 32.3 at 120k on the	
161	reported configuration, all within a band of 1.5. We report	
162	80k.	
163	Capacity, and why we give the whole curve. A 2.2×	
164	larger network (161.2M against 72.5M) leads early and loses	
165	late: 46.1 against 54.2 at 20k steps, 38.3 against 35.8 at 40k,	
166	and 38.5 against 31.8 at 80k. Read at 20k it is a clear win;	
167	read at 80k it is a clear loss. On plain CFG the larger model	
168	is marginally better throughout (114.5 against 117.9), so the	
169	reversal is specific to the configuration we report. The ex-	
170	planation we find plausible is that 2.2 times the parameters	
171	overfit a 50k-sprite corpus sooner, but we did not test it, and	
172	the reason we show three points rather than one is that a	
173	single point would have supported either conclusion — the	
174	same trap that the corpus experiment in Section 3 fell into.	
175	6. Resolutions outside the target range	
176	The body covers 12–24 px, the sizes the medium is actually	
177	used at. One rung above, at 32 px, the ordering is the same:	
178	CFG 1.5 scores 139.2 native FD, ICG $w=2$ 97.1, and ICG	
179	with an eight-colour projection 74.5 (twelve colours: 78.7).	
180	The best k is 8, 4, 8, 6 and 8 at 12, 16, 20, 24 and 32 px:	
181	it neither rises with the canvas nor tracks the real median	
182	palette, and 16 px — the size we study most — is the one	
183	that wants the fewest colours. Absolute values are not com-	
184	parable across resolutions: each has its own reference set	
185	and its own real-versus-real floor.	
186	7. Alternative reference sets	
187	Object-only reference. About a fifth of the corpus is not a	
188	whole object: effects, fragments of larger drawings, glyphs.	
189	Restricting the reference to the 825 held-out sprites a clas-	
190	sifier calls objects, and the prompts to the matching 2,181,	
191	gives 37.1 for the reported configuration. The ordering of	
	methods is unchanged; the absolute value moves because	192
	the reference population did.	193
	Clean subset. Removing the 41.4% of held-out prompts	194
	that have a near-duplicate twin in training leaves 1,759	195
	prompts and moves the reported configuration from 31.8 to	196
	29.0, with every other row moving by at most 3.0 and no	197
	ordering changing.	198
	A second feature space. Inception-v3 clean-FID and KID	199
	on the same samples and the same native reference preserve	200
	the ordering, controls included: 23.79 for the projection,	201
	23.28 at $k=6$, 26.27 without it, 29.37 for CFG, 36.05 for one	202
	global palette, 66.63 for the median filter, 10.54 for the real	203
	held-out sprites. The separation shrinks sharply — a 2.8×	204
	gap between projecting and not projecting under DINOv2	205
	is 1.10× here, and $k=4$ and $k=6$ swap — which is what an	206
	encoder whose input is resized to 299 px would do to 16 px	207
	hard edges. We read it as a check on the ranking and not as	208
	confirmation of the effect size.	209
	8. Prompt alignment in full	210
	CLIP ViT-B/32 image–text similarity (100 cos) and retrieval	211
	of the own caption among 100 distractors, 3,000 held-out	212
	captions, 16 px unless stated. ICG $w=2$ alone: 27.60 /	213
	22.8%. Projection at $k=6$: 27.38 / 22.0%. At $k=4$: 27.07	214
	/ 19.2%. The real held-out sprites the captions were written	215
	from: 27.66 / 22.6%, so at $k=6$ our samples are as retriev-	216
	able from their captions as real sprites are. At 20 px the pro-	217
	jection costs 1.9 points (26.7% to 24.8%) and at 24 px 1.9	218
	points (27.6% to 25.7%).	219
	Two cautions about this measurement. The distractor	220
	draw is re-seeded per directory: with a single shared stream,	221
	a set’s retrieval score depended on which other sets were	222
	scored in the same invocation, which cost us about a point	223
	in an earlier version of these numbers. And a real-sprite	224
	baseline is only meaningful if the sprites are paired with the	225
	captions that describe them — an unpaired reference scores	226
	19.68 / 1.1%, chance by construction, and reading that as	227
	evidence that real pixel art is hard to retrieve would be a	228
	mistake.	229
	9. Licences and release	230
	The sprites added in this work carry per-file records of	231
	archive, member file and licence (CC-BY-4.0, CC-BY-SA-	232
	3.0, OGA-BY-3.0 for the OpenGameArt material; CC0-1.0	233
	for the Kenney packs used in the data ablation). The older	234
	material predates that bookkeeping; we release the manifests	235
	for what we added, together with the scripts that rebuild the	236
	rest from public archives and a table mapping every number	237
	in the paper to the script that produces it.	238